

Event Performance Analysis

Player Behavior & Monetization

Data Analyst Case Study

Prepared by: Vesela Vekieva
March 2026

Contents

[Objective of the Analysis](#)

[Dataset Description](#)

[Data Preparation](#)

[Analysis Approach and Tools](#)

[Other Possible Analytical Tools](#)

[Player Drop-off by Level Group](#)

[Revenue by Level Group](#)

[Revenue per Skip by Level Group](#)

[Drop-off vs Revenue per Skip by Level Group](#)

[Top 10 Missions by Player Drop-off](#)

[Top 10 Missions by Revenue per Skip](#)

[Total Event Revenue](#)

[Key Findings](#)

[Recommendations for Future Events](#)

[A/B Test Analysis – Monetization Feature](#)

[A/B Test Interpretation](#)

Objective of the Analysis

This analysis evaluates **player behavior** and **monetization performance** during a special in-game event.

The analysis focuses on two main areas:

- **Churn Analysis**
 - Which missions cause the highest player drop-off?
 - Do players behave differently depending on their level?
- **Revenue Analysis**
 - How much revenue was generated during the event?
 - Which player levels generate the most revenue?

The goal is to identify **player friction points** and **monetization opportunities** that can improve future events.

Dataset Description

The dataset contains **mission progression data** for players participating in the event.

- The dataset includes **2040 rows**, each representing a player mission.
- For each mission and player level, the dataset includes:
 - *Player level;*
 - *Mission name;*
 - *Number of players starting the mission;*
 - *Number of missions skipped;*
 - *Premium currency cost to skip the mission;*
- The premium currency conversion rate is **100 currency = \$4.99**.

Since players can skip missions using premium currency, skip behavior provides insight into both **player frustration points** and **monetization patterns**.

Data Preparation

Before running the analysis, the dataset was structured in Excel and additional metrics were created to better evaluate player behavior and monetization.

- **Completion Rate:** This metric shows the percentage of players who successfully progress through a mission and helps identify potential difficulty spikes.
- **Revenue per Mission:** This metric estimates the revenue generated from mission skips.
- **Revenue per Skip:** This metric helps evaluate how much revenue is generated from each skipped mission.
- Pivot tables were used to aggregate results by level groups and mission.
- Player levels were also **grouped into 5 level ranges** to analyze behavioral patterns across progression stages.

These transformations allowed the dataset to be used for both **player behavior analysis** and **revenue analysis**.

Analysis Approach and Tools

The analysis was conducted using **Microsoft Excel**, which is suitable for structured exploratory analysis and visualization of medium-sized datasets.

The workflow included:

- Organizing the dataset as an Excel table.
- Creating calculated metrics:
 - *Completion Rate*;
 - *Revenue per Mission*;
 - *Revenue per Skip*;
- Using **Pivot Tables** to aggregate results by:
 - *Mission*;
 - *Player level groups*;
- Creating visualizations to highlight **player drop-off** and **monetization patterns**.

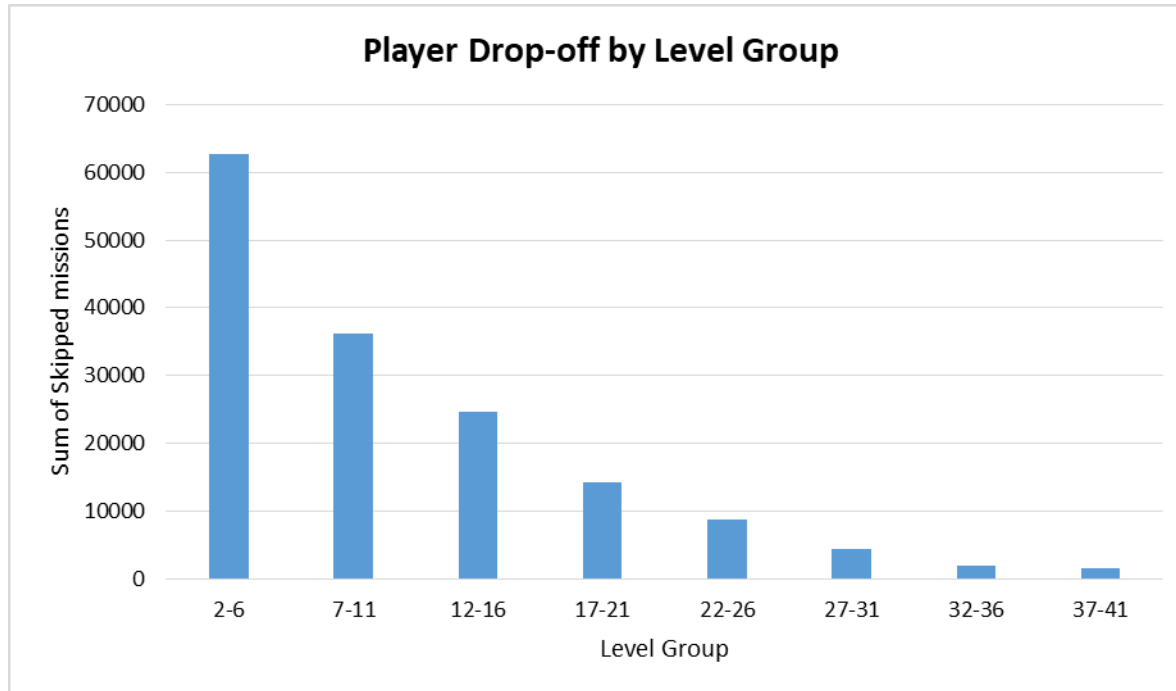
Other Possible Analytical Tools

While Excel was sufficient for this analysis, other tools could also be used depending on dataset size and complexity.

- **Structured Query Language (SQL)**
 - Suitable for querying large datasets stored in databases.
 - Efficient for filtering and aggregating millions of records.
- **Python**
 - Useful for advanced statistical analysis, custom visualizations and automation of repetitive tasks.
 - Allows machine learning models and deeper behavioral analysis.
- **Business Intelligence tools (Tableau / Power BI)**
 - Suitable for interactive dashboards.
 - Useful for continuous monitoring of event performance.

For this dataset (~2000 rows), Excel provided a **fast and transparent analytical workflow**.

Player Drop-off by Level Group



Insight: Player drop-off is concentrated in the **early progression stages**, suggesting that many players encounter friction early in the event and fail to progress to later missions.

What was done:

Player levels were grouped into ranges of five to analyze player behavior across different stages of progression.

The number of **skipped missions** was aggregated for each level group using a pivot table.

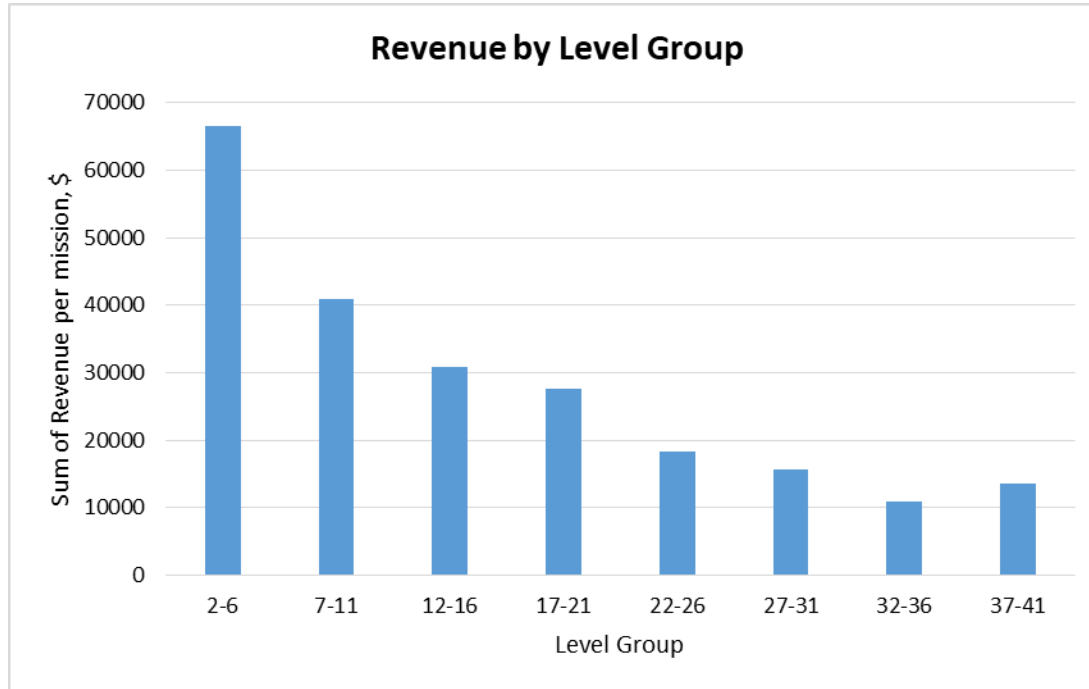
This allows us to identify **where players are most likely to drop off during the event progression**.

Key Findings:

The highest number of skipped missions occurs in the **early level groups (2–6 and 7–11)**.

As player level increases, the number of skipped missions **declines significantly**, indicating that fewer players reach later stages of the event.

Revenue by Level Group



Insight: Although early levels generate the highest total revenue, this is primarily driven by **player volume rather than higher spending per player**.

What was done:

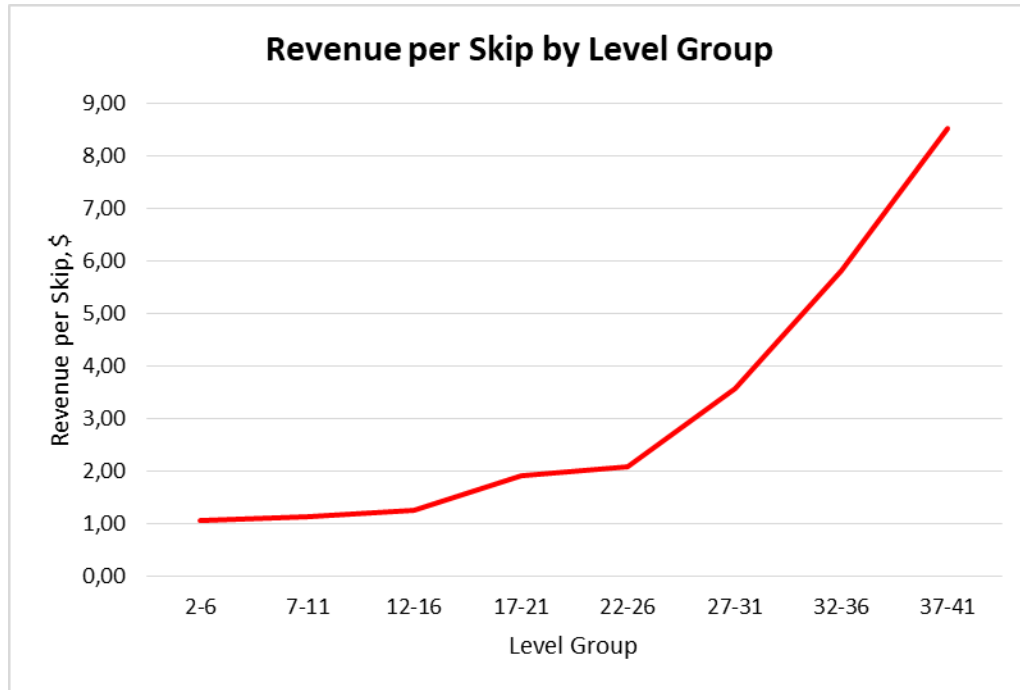
Revenue generated from mission skips was calculated and aggregated for each level group. This shows how much revenue is generated at different stages of player progression.

Key Findings:

The highest total revenue is generated in the **early level groups**, where the player population is largest.

Revenue gradually decreases as player levels increase, reflecting the decline in the number of players progressing through the event.

Revenue per Skip by Level Group



Insight: Players who progress to later stages appear **more willing to spend premium currency**, making late-game missions highly valuable from a monetization perspective.

What was done:

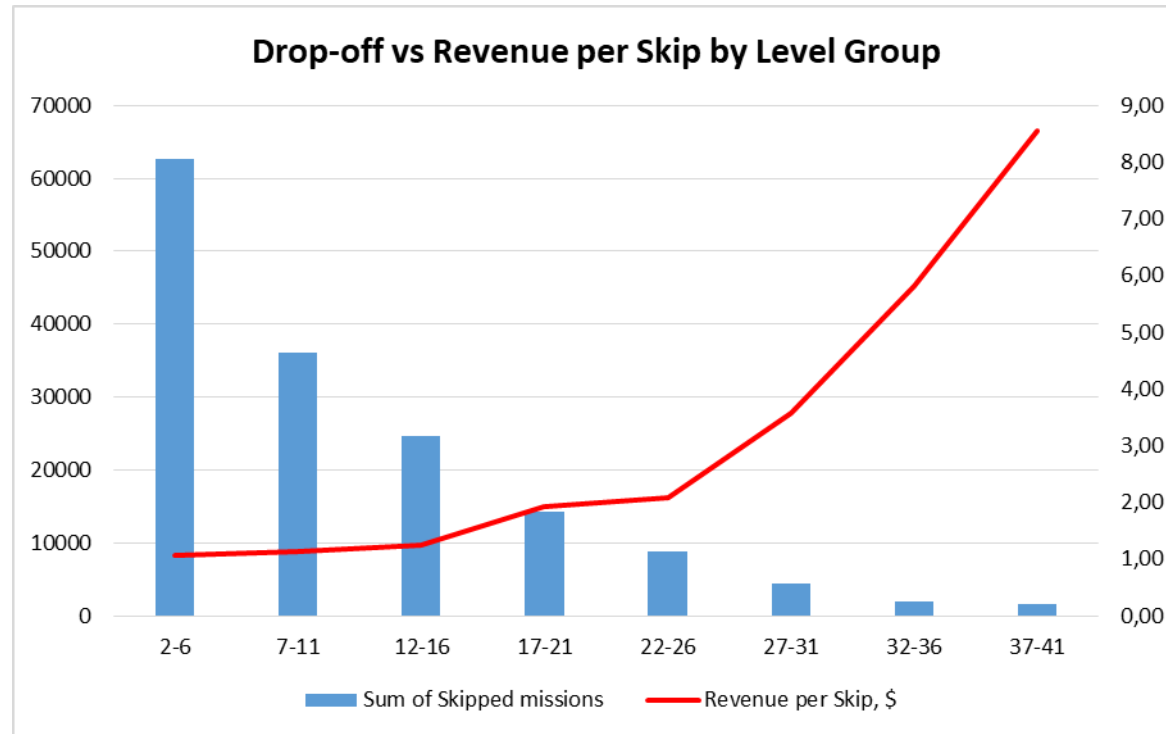
Revenue per Skip was calculated to understand how much revenue is generated on average from each skipped mission.

This metric highlights monetization efficiency across different stages of player progression.

Key Findings:

Revenue per Skip increases steadily with player level. While early levels generate around **\$1 per skip**, later levels reach values above **\$8 per skip**.

Drop-off vs Revenue per Skip by Level Group



Insight: Later stages of the event generate **higher value per player**, even though fewer players reach them.

What was done:

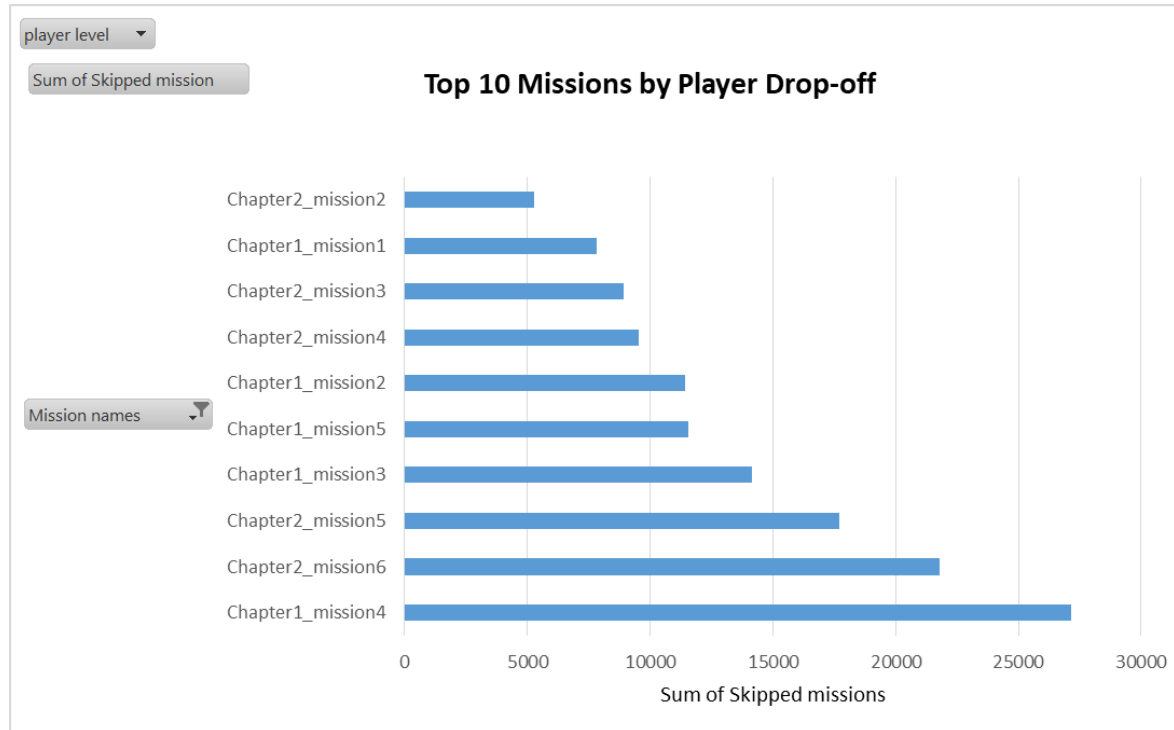
A combined chart was created to compare **player drop-off** and **revenue per skip** across level groups.

This visualization highlights the relationship between **player retention** and **monetization potential**.

Key Findings:

Early levels show **high drop-off but low revenue per skip**, while later levels show **lower player counts but significantly higher revenue per skip**.

Top 10 Missions by Player Drop-off



Insight: Early event missions appear to create the strongest friction for players, suggesting that difficulty spikes or mission design in early chapters may disrupt player progression.

What was done:

A pivot table was used to aggregate the total number of **skipped missions by mission name**.

The missions were then sorted in descending order to identify the **top 10 missions with the highest player drop-off**. This analysis helps identify specific missions where players are most likely to skip or abandon progression.

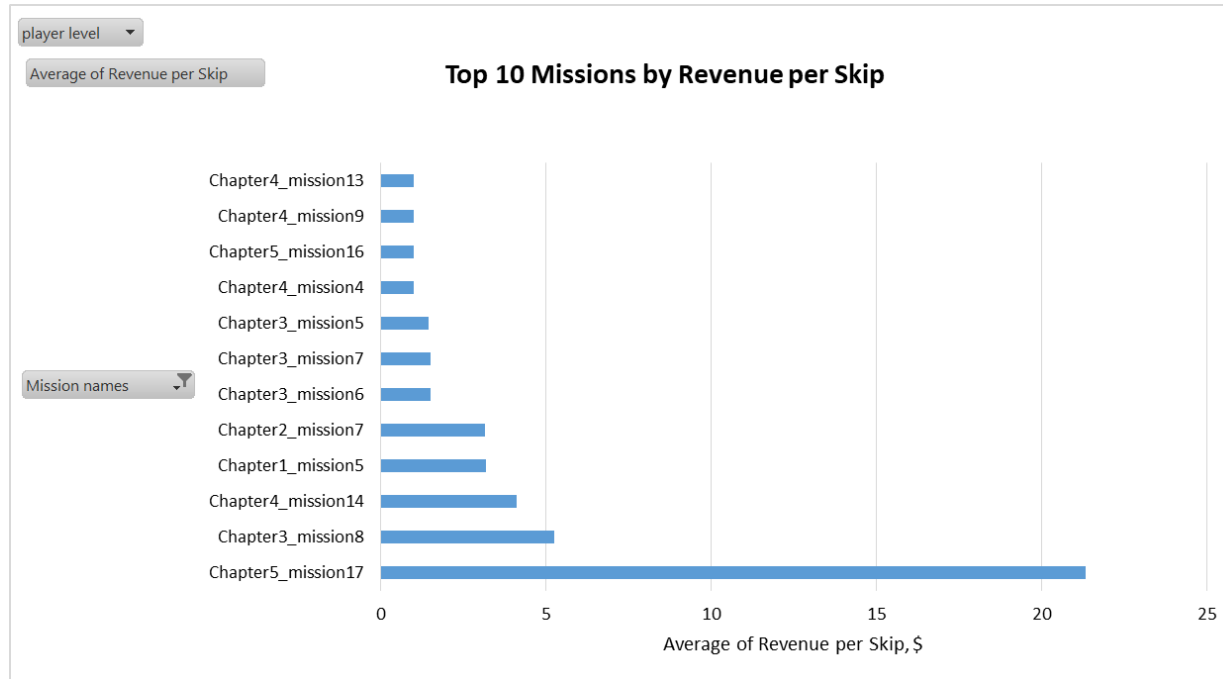
Key Findings:

The highest player drop-off occurs in early missions, particularly:

- *Chapter1_mission4*
- *Chapter2_mission6*
- *Chapter2_mission5*

These missions account for a large share of the total skipped missions during the event.

Top 10 Missions by Revenue per Skip



What was done:

Revenue per Skip was calculated for each mission to evaluate the **monetization efficiency of skipped missions**.

A pivot table was used to identify the missions generating the **highest revenue per skip**.

Key Findings:

The mission **Chapter5_mission17** stands out with a significantly higher **Revenue per Skip of 21.33**, which is much higher than the other missions.

Further inspection shows that this mission has the **highest skip cost (900 premium currency)** in the dataset, while most other missions have skip costs ranging between **15 and 135 premium currency**.

As a result, this mission generates **much higher revenue per skipped mission** compared to the others.

Insight: Later-stage missions tend to generate **higher revenue per skip**, suggesting that players who progress further in the event may be more willing to spend premium currency to continue playing.

However, the unusually high revenue per skip for **Chapter5_mission17** appears to be primarily driven by its **higher skip cost**, rather than higher player skip frequency.

This indicates that **pricing design plays an important role in monetization outcomes**, and highlights the need to balance mission difficulty and skip pricing.

Total Event Revenue

Row Labels	Sum of Revenue per mission
2-6	66463,307
7-11	40852,3815
12-16	30810,0065
17-21	27533,323
22-26	18334,5075
27-31	15580,0275
37-41	13554,337
32-36	10983,489
Grand Total	224111,379

Total revenue generated during the event was calculated by summing the revenue generated from all skipped missions.

Total Event Revenue:
\$ 224,111

***Insight:** Although most revenue is generated in early levels due to higher player volume, later levels generate **higher revenue per player**, highlighting the importance of retaining players deeper into the event progression.*

Key Findings

Key Findings from the Event Analysis

- Player drop-off is concentrated in the **early level groups (2–11)**, indicating that many players encounter friction early in the event progression.
- The missions with the highest drop-off are primarily located in the **early chapters**, suggesting possible difficulty spikes or progression barriers.
- Early levels generate the **highest total revenue**, mainly due to the larger number of players reaching these stages.
- Later levels generate **significantly higher revenue per skip**, meaning that players who progress further tend to spend more premium currency.
- Some missions generate unusually high revenue due to **higher skip costs**, showing that pricing design has a strong impact on monetization outcomes.
- Overall event revenue reached **\$ 224,111**, indicating that mission skips represent a significant monetization mechanism during the event.

Recommendations for Future Events

Based on the analysis, several improvements could increase both participation and revenue:

- **Improve early progression**
 - Reduce early difficulty spikes or improve mission clarity to reduce player drop-off in early levels.
- **Optimize mission difficulty curve**
 - Gradually increase difficulty to maintain engagement while avoiding frustration.
- **Leverage late-game monetization**
 - Since players in later stages generate higher revenue per skip, events should encourage deeper progression.
- **Balance skip pricing**
 - High skip prices can significantly increase revenue, but should be balanced carefully to avoid discouraging players.
- **Use mission design to guide spending**
 - Introduce optional challenges, milestone rewards, or time-limited objectives that encourage players to progress further in the event.

A/B Test Analysis – Monetization Feature

An A/B test was conducted on a monetization feature to evaluate its impact on player spending behavior.

Test Results

Metric	Group A (Control)	Group B
ARPDau \$	0.15	0.16
ARPPU \$	30	16

ARPDau - Average Revenue Per Daily Active User

ARPPU - Average Revenue Per Paying User

Observation:

Group B generates **higher ARPDau**, while Group A has **significantly higher ARPPU**.

A/B Test Interpretation

Which scenario wins?

Preferred Scenario: Group B

Although Group A generates higher revenue per paying user, Group B produces **higher overall revenue per active user (ARPDau)**.

This suggests that the tested feature likely **encouraged more players to spend**, even if individual spending amounts were lower.

Since ARPDau reflects **total monetization performance**, Group B would likely generate higher overall revenue at scale.

A/B Test Interpretation

Why do these metrics behave this way?

The difference between ARPDau and ARPPU suggests that the feature likely **increased the number of paying players**, but reduced the average spending per payer.

Possible reasons include:

- *Lower-priced offers*
- *Easier access to premium currency*
- *More frequent microtransactions*

This pattern often indicates that monetization became more accessible to a wider player base, increasing the number of paying users while reducing the average spend per payer.

A/B Test Interpretation

What feature could have been tested?

The feature tested could include:

- Introducing ***lower-priced offers***
- Adding ***small premium currency bundles***
- Introducing ***limited-time microtransactions***
- Adding ***discounted mission skips***

These features typically increase the number of paying users, which raises ARPDAU while lowering ARPPU.